

OPEN MOCK TEST - 2 (Code-A)

Test Date : 19/06/2022

ANSWERS

1. (1)	41. (1)	81. (3)	121. (3)	161. (4)
2. (3)	42. (3)	82. (4)	122. (4)	162. (4)
3. (2)	43. (2)	83. (4)	123. (3)	163. (3)
4. (4)	44. (2)	84. (3)	124. (2)	164. (4)
5. (1)	45. (1)	85. (1)	125. (2)	165. (3)
6. (2)	46. (1)	86. (4)	126. (3)	166. (2)
7. (2)	47. (3)	87. (3)	127. (3)	167. (3)
8. (3)	48. (4)	88. (3)	128. (4)	168. (4)
9. (4)	49. (2)	89. (4)	129. (2)	169. (3)
10. (1)	50. (1)	90. (2)	130. (2)	170. (2)
11. (4)	51. (3)	91. (2)	131. (3)	171. (3)
12. (3)	52. (2)	92. (3)	132. (4)	172. (2)
13. (1)	53. (2)	93. (2)	133. (3)	173. (4)
14. (3)	54. (1)	94. (3)	134. (2)	174. (2)
15. (4)	55. (1)	95. (2)	135. (4)	175. (3)
16. (2)	56. (3)	96. (2)	136. (3)	176. (3)
17. (3)	57. (3)	97. (2)	137. (2)	177. (3)
18. (2)	58. (2)	98. (3)	138. (4)	178. (1)
19. (4)	59. (3)	99. (1)	139. (1)	179. (2)
20. (1)	60. (3)	100. (1)	140. (2)	180. (3)
21. (2)	61. (1)	101. (3)	141. (2)	181. (2)
22. (4)	62. (3)	102. (1)	142. (4)	182. (2)
23. (1)	63. (3)	103. (3)	143. (3)	183. (2)
24. (1)	64. (2)	104. (3)	144. (1)	184. (1)
25. (4)	65. (4)	105. (4)	145. (2)	185. (4)
26. (2)	66. (3)	106. (2)	146. (2)	186. (3)
27. (2)	67. (1)	107. (2)	147. (2)	187. (4)
28. (4)	68. (2)	108. (3)	148. (1)	188. (2)
29. (3)	69. (1)	109. (4)	149. (2)	189. (3)
30. (4)	70. (2)	110. (2)	150. (4)	190. (3)
31. (2)	71. (4)	111. (2)	151. (2)	191. (1)
32. (1)	72. (2)	112. (3)	152. (3)	192. (3)
33. (4)	73. (4)	113. (1)	153. (4)	193. (2)
34. (1)	74. (3)	114. (3)	154. (1)	194. (2)
35. (3)	75. (3)	115. (3)	155. (2)	195. (3)
36. (1)	76. (2)	116. (3)	156. (3)	196. (3)
37. (2)	77. (1)	117. (2)	157. (3)	197. (1)
38. (4)	78. (4)	118. (1)	158. (3)	198. (3)
39. (3)	79. (3)	119. (3)	159. (2)	199. (1)
40. (2)	80. (3)	120. (2)	160. (2)	200. (2)

HINTS & SOLUTIONS

[PHYSICS]

SECTION-A

1. Answer (1)

Hint: $\frac{\Delta V}{V} = 3 \times \frac{\Delta r}{r}$

Sol.: $V = \frac{4}{3}\pi r^3$

$\frac{\Delta V}{V} = 3 \frac{\Delta r}{r}$

$\frac{\Delta V}{V} \times 100 = 3 \times 1.5 = 4.5\%$

2. Answer (3)

Hint: Galileo law of odd numbers.

Sol.: $h_1 : h_2 : h_3 = 1 : 3 : 5$

$\Rightarrow h_1 = x, h_2 = 3x, h_3 = 5x$

Given, $h_1 + h_2 = h \Rightarrow 4x = h \Rightarrow x = \frac{h}{4}$

$\Rightarrow h_3 = 5x = \frac{5h}{4} = 1.25h$

3. Answer (2)

Hint: $a_x = \frac{d^2x}{dt^2}$ and $a_y = \frac{d^2y}{dt^2}$

Sol.: $a_x = \frac{d^2(4t^2)}{dt^2} = \frac{d}{dt} \left(\frac{d(4t^2)}{dt} \right) = 8 \text{ m/s}^2$

$a_y = \frac{d^2(8t)}{dt^2} = \frac{d}{dt} \left(\frac{d(8t)}{dt} \right) = 0$

So, $a = 8 \text{ m/s}^2$

4. Answer (4)

Hint & Sol: A vector can be resolved into infinite number of components.

5. Answer (1)

Hint: $T = 2\pi \sqrt{\frac{l}{g_{\text{eff}}}}$

Sol.: Case 1: $T = 2\pi \sqrt{\frac{l}{g}}$

Case 2: $T' = 2\pi \sqrt{\frac{l}{g - g/4}} = \frac{2}{\sqrt{3}} \left(2\pi \sqrt{\frac{l}{g}} \right)$

$\therefore T' = \frac{2}{\sqrt{3}} T = \frac{2}{\sqrt{3}} \times 6 = 4\sqrt{3} \text{ s}$
 $= 6.93 \text{ s}$

6. Answer (2)

Hint: At centre of mass, moment of inertia is minimum.

Sol.: $I = 8x^2 - 4x + 2$

For minimum moment of inertia,

$\frac{dI}{dx} = 16x - 4 = 0$

$\Rightarrow x = \frac{1}{4} = 0.25 \text{ m}$

7. Answer (2)

Hint: Parallel and series combination of capacitors.

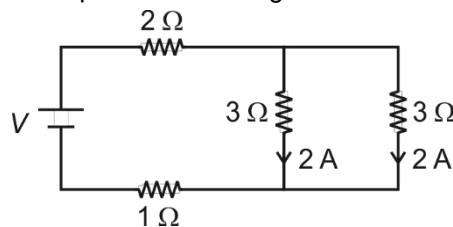
Sol.: Three $3 \mu\text{F}$ capacitors are in series and their equivalent combination is in parallel to $1 \mu\text{F}$ capacitor

$C_{\text{net}} = \frac{3}{3} + 1 = 2 \mu\text{F}$

8. Answer (3)

Hint: Use current division rule in parallel combination of resistors

Sol.: Simplified circuit diagram.



$i_{\text{net}} = 2 \text{ A} + 2 \text{ A} = 4 \text{ A}$

$R_{\text{net}} = 2 + 1 + \frac{3}{2} = 4.5 \Omega$

$V = i_{\text{net}} R_{\text{net}} = 4 \times 4.5 = 18 \text{ V}$

9. Answer (4)

Hint: $E_n = -\frac{13.6}{n^2} \text{ eV}$ and $L = \frac{nh}{2\pi}$

Sol.: $-3.4 = \frac{-13.6}{n^2} \Rightarrow n^2 = 4 \Rightarrow n = 2$

$\Rightarrow L = \frac{nh}{2\pi} = \frac{h}{\pi} = \frac{3h}{3\pi}$

10. Answer (1)

Hint & Sol.: $P = \frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$

- R_1 and R_2 remains unchanged when cut along principal axis.
- When biconvex lens is cut perpendicular to principal axis radius of curvature becomes infinity on the surface which is cut.

11. Answer (4)

Hint & Sol.: In case of motion of charged particle in magnetic field, kinetic energy remain same. The velocity and linear momentum, however keeps on changing.

12. Answer (3)

Hint: $g' = \frac{g}{(1 + h/R)^2}$

Sol.: At height $h = R$

$$g' = \frac{g}{(1 + R/R)^2} = \frac{g}{4}$$

$$\frac{mg'}{mg} = \frac{1}{4} \Rightarrow mg = 4 \times mg'$$

$$= 960 \text{ N}$$

13. Answer (1)

Hint & Sol.: For each degree of freedom, average thermal kinetic energy is equal to $\frac{1}{2} k_B T$ (irrespective of atomicity)

14. Answer (3)

Hint: $\tan \delta = \frac{B_V}{B_H}$

Sol.: $\tan(37^\circ) = \frac{B_V}{0.2} \Rightarrow B_V = 0.2 \times \frac{3}{4}$

$$\Rightarrow B_V = \frac{3}{20} = 0.15 \text{ G}$$

15. Answer (4)

Hint & Sol.: This is an OR gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

16. Answer (2)

Hint: $i_{\text{rms}} = \frac{V_{\text{rms}}}{Z}$

Sol.: $Z = \sqrt{R^2 + (X_L - X_C)^2}$

$$Z = \sqrt{2^2 + (4 - 2)^2} = 2\sqrt{2} \Omega$$

$$\Rightarrow i_{\text{rms}} = \frac{220/\sqrt{2}}{2\sqrt{2}} = \frac{220}{4} = 55 \text{ A}$$

17. Answer (3)

Hint: $\langle u \rangle = \frac{1}{2} \epsilon_0 E_{\text{rms}}^2$

Sol.: $\langle u \rangle = \frac{1}{2} \epsilon_0 \left(\frac{20}{\sqrt{2}} \right)^2$

$$= \frac{8.85 \times 10^{-12}}{2} \times 200$$

$$= 8.85 \times 10^{-10} \text{ J/m}^3$$

18. Answer (2)

Hint: $a = \mu g$

Sol.: $a = \mu g = 0.2 \times g = 2 \text{ m/s}^2$

Final speed of block = 10 m/s

$$10 = 2 \times t \Rightarrow t = 5 \text{ s}$$

Distance travelled w.r.t. ground

$$= \frac{1}{2} at^2 = \frac{1}{2} \times (2)(5)^2 = 25 \text{ m}$$

19. Answer (4)

Hint & Sol.: Inside a uniformly charged spherical shell, the electric field is zero in electrostatic condition.

20. Answer (1)

Hint: $\mu = \frac{\sin \left(\frac{A + \delta_m}{2} \right)}{\sin \left(\frac{A}{2} \right)}$

Sol.: $\sqrt{2} = \frac{\sin \left(\frac{60^\circ + \delta_m}{2} \right)}{\sin (60^\circ/2)}$

On solving, $\delta_m = 30^\circ$

Also, $\delta_m = 2i - A = 2i - 60^\circ = 30^\circ$

$$\Rightarrow i = 45^\circ$$

21. Answer (2)

Hint: $y_{\max} = \frac{n\lambda D}{d}$

$$y_{\min} = (n - 1/2) \frac{\lambda D}{d}$$

Sol.: $y_{I\max} = \frac{\lambda D}{d}$

$$y_{III\min} = \left(3 - \frac{1}{2}\right) \frac{\lambda D}{d} = \frac{5\lambda D}{2d}$$

$$\Delta y = \frac{5\lambda D}{2d} - \frac{\lambda D}{d} = \frac{3\lambda D}{2d}$$

$$\Delta y' = \frac{3 \times \lambda (2D)}{2(d/2)} = 4 \left(\frac{3\lambda D}{2d} \right) = 4\Delta y$$

22. Answer (4)

Hint: Power = $\frac{dW}{dt}$

Sol.: $P = \frac{dW}{dt} = \frac{d(4t^2)}{dt}$

$$\Rightarrow P = 8t$$

$$\text{At } t = 2 \text{ s, } P = 16 \text{ W}$$

23. Answer (1)

Hint: $I = \frac{P}{\text{area}} = \frac{P}{4\pi r^2}$

Sol.: $I = \frac{16\pi}{4\pi(100)^2}$
 $= 4 \times 10^{-4} \text{ W/m}^2$

24. Answer (1)

Hint: T.E. = $-13.6 \frac{Z^2}{n^2}$

Sol.: For 1st excited state of singly ionised He-atom

$$(n = 2, Z = 2)$$

$$\Rightarrow \text{T.E.} = -13.6 \text{ eV} = E$$

$$\text{Also T.E.} = \frac{U}{2} = -\text{K.E.}$$

$$\Rightarrow \text{K.E.} = -E \text{ and } U = 2E$$

25. Answer (4)

Hint: Use equation of continuity

Sol.: $A_1V_1 + A_2V_2 = A_3V_3$

$$\Rightarrow 2A + 5 \times (2A) = v(3A)$$

$$\Rightarrow v = 4 \text{ m/s}$$

26. Answer (2)

Hint: As per Kepler's law $T^2 \propto R^3$

Sol.: $T^2 \propto R^3$

$$T^2 \propto (2R)^3$$

$$\Rightarrow \left(\frac{T'}{T}\right) = (2)^{3/2} = 2\sqrt{2}$$

27. Answer (2)

Hint & Sol.: For adiabatic compression

$$\Delta Q = 0, \Delta W < 0$$

$$\text{Also } \Delta Q = \Delta U + \Delta W$$

$$\Rightarrow \Delta U = -\Delta W > 0$$

28. Answer (4)

Hint: $\frac{\Delta Q}{\Delta t} = \frac{kA\Delta T}{L}$

Sol.: $\left(\frac{\Delta Q}{\Delta t}\right)_{C \rightarrow A} = \left(\frac{\Delta Q}{\Delta t}\right)_{B \rightarrow C}$

$$\Rightarrow \frac{kA(T_C - 20)}{3L} = \frac{kA(100 - T_C)}{L}$$

$$\Rightarrow T_C = 80^\circ\text{C}$$

29. Answer (3)

Hint: $Y = \frac{(F/A)}{(\Delta l/l)}$

$$\text{Energy} = \frac{1}{2} \text{ stress} \times \text{strain} \times \text{volume}$$

Sol.: $Y = \frac{\text{Stress}}{\text{Strain}} \Rightarrow \text{Strain} = \frac{\text{Stress}}{Y} = \frac{Mg}{AY}$

$$\Rightarrow \text{Energy} = \frac{1}{2} \times \frac{Mg}{A} \times \frac{Mg}{AY} \times Al$$

$$= \frac{M^2 g^2 l}{2AY}$$

30. Answer (4)

Hint & Sol.: 29.68 $\rightarrow$ as digit to be dropped is greater than five. Thus the preceding digit is increased by one.

3.762 $\rightarrow$ as digit to be dropped is less than five. Thus the preceding digit is left unchanged.

So numbers are reported as 29.7 and 3.76

31. Answer (2)

Hint: $|\vec{A} \cdot \vec{B}| = AB \cos \theta$

$$|\vec{A} \times \vec{B}| = AB \sin \theta$$

Sol.: $\cos \theta = \frac{|\vec{A} \cdot \vec{B}|}{AB}$

$\sin \theta = \frac{|\vec{A} \times \vec{B}|}{AB}$

So, given expression is

$\sin^2 \theta + \cos^2 \theta = 1$

32. Answer (1)

Hint: For closed organ pipe

$f_{n^{\text{th}} \text{ harmonic}} = \frac{n\nu}{4l}$

For open organ pipe

$f_{n^{\text{th}} \text{ harmonic}} = \frac{n\nu}{2l}$

Sol.: Second overtone of open pipe is equal to third harmonic

so, $\frac{\nu}{4l} = \frac{3\nu}{2l_0}$

$\Rightarrow l_0 = 6l$

33. Answer (4)

Hint: $N = \frac{N_0}{2^n}$

n : number of half lives

Sol.: $N = \frac{N_0}{2^3} = \frac{N_0}{8}$

Decayed fraction : $\frac{N_0 - N}{N_0}$

$= \frac{N_0 - \frac{N_0}{8}}{N_0}$

$= \frac{7}{8}$

34. Answer (1)

Hint & Sol.: The band gap in germanium and silicon in eV are 0.7 and 1.1 respectively.

35. Answer (3)

Hint & Sol.: Power = Intensity $\times$ Area
 $= 15 \times 4 = 60 \text{ W}$

SECTION-B

36. Answer (1)

Hint & Sol.: From color coding of carbon resistors
 Red $\rightarrow 2$

Green $\rightarrow 5$

Blue $\rightarrow 6$

Silver $\rightarrow 10\%$

So, $R = 25 \times 10^6 \Omega \pm 10\%$
 $= 2.5 \times 10^7 \Omega \pm 10\%$

37. Answer (2)

Hint: For electron $\lambda = \frac{12.27}{\sqrt{V}} \text{ \AA}$

Sol.: Initial Energy of electron is 120 eV, i.e. accelerated through a potential difference of 120 V.

$\lambda = \frac{12.27}{\sqrt{120 + 49}} = \frac{12.27}{\sqrt{169}}$
 $= \frac{12.27}{13} = 0.94 \text{ \AA}$

38. Answer (4)

Hint & Sol.: For a lens, $m = \frac{f}{f + u}$

$\Rightarrow -2 = \frac{10}{10 + u}$

$\Rightarrow 10 + u = -5 \Rightarrow u = -15 \text{ cm}$

39. Answer (3)

Hint: $|e| = \left| \frac{L di}{dt} \right|$

Sol.: $2 = L \times \frac{(6 - (-2))}{10}$

$\Rightarrow 2 = L \times \frac{8}{10}$

$\Rightarrow L = \frac{20}{8} = 2.5 \text{ H}$

40. Answer (2)

Hint & Sol.: $T = \frac{2\pi m}{qB}$

Thus $T \propto \nu^0$

41. Answer (1)

Hint: Series and parallel combination of resistances

Sol.: $R_{AB} = \frac{2 \times 6}{2 + 6} = \frac{12}{8} = \frac{3}{2} \Omega$

$R_{AC} = \frac{4 \times 4}{4 + 4} = 2 \Omega$

$\therefore \frac{R_{AB}}{R_{AC}} = \frac{3}{2} : 2 = 3 : 4$

42. Answer (3)

Hint: $E = \frac{kQ}{r^2}$

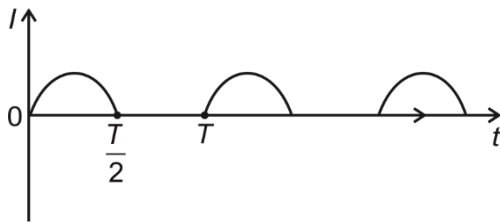
Sol.: $E = \frac{9 \times 10^9 \times Q}{0.4^2} = 4$

$\Rightarrow Q = 7.11 \times 10^{-11} \text{ C}$

43. Answer (2)

Hint & Sol.

- Diode and half wave rectifier will trim half of the current



$$I_{\text{rms}} \left(0 \text{ to } \frac{T}{2} \right) = \frac{I_0}{\sqrt{2}}$$

$$I_{\text{rms}} \left(\frac{T}{2} \text{ to } T \right) = 0$$

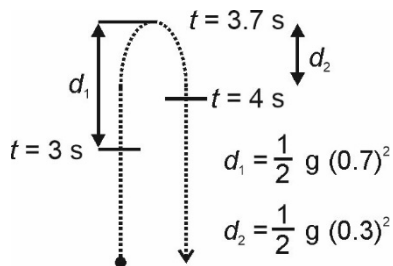
$$\Rightarrow I_{\text{rms}}(0 \text{ to } T) = \sqrt{\frac{\left(\frac{I_0}{\sqrt{2}} \right)^2 + 0^2}{2}} = \frac{I_0}{2}$$

- Full wave rectifier and resistor will not trim the current so, $I_{\text{rms}} = \frac{I_0}{\sqrt{2}}$.

44. Answer (2)

Hint: Motion under gravity

Sol.:



$$d_{\text{net}} = d_1 + d_2 = \frac{1}{2} g (0.7^2 + 0.3^2) = 2.9 \text{ m}$$

45. Answer (1)

Hint: $v = \frac{dx}{dt}$, $a = \frac{dv}{dt}$

$F = ma$

Sol.: $v = \frac{dx}{dt} = 3t^2 - 24$

Now, $a = \frac{dv}{dt} = 6t$

$F = ma = 4 \times 6t = 24t$

At $t = 2 \text{ s}$

$F = 24 \times 2 = 48 \text{ N}$

46. Answer (1)

Hint: $W = \int \vec{F} \cdot d\vec{r}$

Sol.: $d\vec{r} = dx\hat{i} + dy\hat{j}$

$$W = \int_{(1,2)}^{(0,2)} (xy^2 dx + yx^2 dy) = \int_{(1,2)}^{(0,2)} \frac{d(x^2 y^2)}{2} = \left[\frac{x^2 y^2}{2} \right]_{1,2}^{0,2} = 0 - 2 = -2 \text{ J}$$

47. Answer (3)

Hint: $I = \frac{2}{5} mR^2$

$$\begin{aligned} \text{Sol.} \quad \frac{I_1}{I_2} &= \frac{\frac{2}{5} m_1 R_1^2}{\frac{2}{5} m_2 R_2^2} \\ &= \frac{m_1 R_1^2}{m_2 R_2^2} \\ &= \frac{\delta \left(\frac{4}{3} \pi R_1^3 \right) R_1^2}{\delta \left(\frac{4}{3} \pi R_2^3 \right) R_2^2} \\ &\Rightarrow \left(\frac{R_1}{R_2} \right)^5 = \frac{1}{x^5} \end{aligned}$$

48. Answer (4)

Hint & Sol.: Vertical height of water column remains same.

$l \sin 37^\circ = h = 6$

$\Rightarrow l = 6 \times \frac{5}{3} = 10 \text{ cm}$

49. Answer (2)

Hint: $P = \sigma AT^4$

$$\text{Sol.: } \frac{P_2}{P_1} = \left(\frac{T_2}{T_1}\right)^4 = \left(\frac{273 + 273}{273}\right)^4$$

$$= 2^4 = 16$$

$$\Rightarrow P_2 = 16P$$

50. Answer (1)

Hint & Sol.: $\Delta v = \text{area of } a-t \text{ graph}$

$$v - 0 = \frac{1}{2} \times 20 \times 2$$

$$v = 20 \text{ m/s}$$

[CHEMISTRY]

SECTION-A

51. Answer (3)

Hint: Ratio of number of atoms in closed packed lattice to tetrahedral voids = 1 : 2.

Sol.: Each HCP unit cell contains 6 atoms therefore, number of tetrahedral voids will be 12.

52. Answer (2)

Hint: Mn^+ has $4s^1 3d^5$ outer electronic configuration.

Sol.: Conversion of Mn^+ into Mn^{2+} involves removal of electron from $4s^1$ therefore exchange energy remains same in Mn^+ and Mn^{2+} ions.

53. Answer (2)

Hint: At constant pressure, $V \propto T$

Sol.: According to Charles' law

$$V = KT \text{ where } K \text{ is proportionality constant} = \frac{nR}{P}$$

Therefore lesser is the pressure, greater is the slope of V-T graph

54. Answer (1)

Hint: Tyndall effect is shown by colloidal solutions.

Sol.: Among the given options except urea all are colloidal solutions therefore urea solution being a true solution does not show Tyndall effect.

55. Answer (1)

Hint: $H = U + PV$

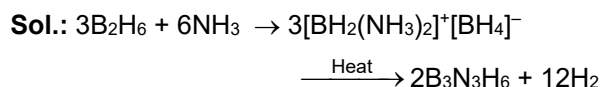
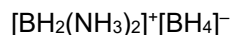
Sol.: $\therefore H = U + PV$

$$H = U + nRT \quad [\because PV = nRT]$$

$$\therefore \Delta H = \Delta U + \Delta n_g RT$$

56. Answer (3)

Hint: $\text{B}_2\text{H}_6 \cdot 2\text{NH}_3$ can be written as

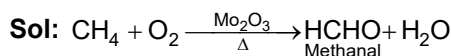


57. Answer (3)

Hint & Sol: Terrestrial hydrogen contains 0.0156% of deuterium in the form of HD

58. Answer (2)

Hint: Alkanes on heating with a regulated supply of dioxygen at high pressure and in presence of suitable catalyst gives various oxidation products.



59. Answer (3)

Hint: At 1200 K, BeCl_2 exists as a linear monomer.

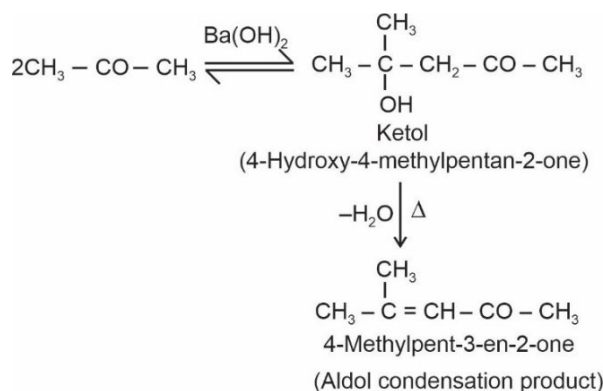
Sol.: In BeCl_2 , Hybridisation of Be is sp and number of electrons around Be is 4



60. Answer (3)

Hint: Methyl ketones in presence of dilute alkali undergoes aldol condensation.

Sol.:



61. Answer (1)

Hint: Cheilosis is fissuring at corners of mouth and lips.

Sol.: Cheilosis is caused due to deficiency of vitamin B₂.

62. Answer (3)

$$\text{Hint: } \Lambda_m^\circ \text{CH}_3\text{COOH} = \frac{1}{2} \Lambda_m^\circ (\text{CH}_3\text{COO})_2\text{Ca} - \frac{1}{2} \Lambda_m^\circ (\text{CaCl}_2) + \Lambda_m^\circ (\text{HCl})$$

Sol.: Using Kohlrausch's Law

$$\Lambda_m^\circ \text{CH}_3\text{COOH} = \frac{200.8}{2} - \frac{271.6}{2} + 426.16 = 390.76 \text{ S cm}^2 \text{ mol}^{-1}$$

63. Answer (3)

Hint: Lanthanoids and actinoids are metallic in nature.

Sol.:

- Due to metallic nature, lanthanoid hydroxides are basic in nature.
- Actinoids are highly reactive metals.
- Due to high reactivity, lanthanoids reacts rapidly in air.
- $\text{Ce}^{4+}/\text{Ce}^{3+}$ can oxidise water but the reaction rate is very slow hence Ce(IV) is a good analytical reagent.

64. Answer (2)

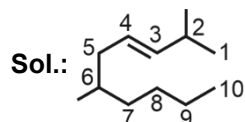
Hint: Cl_2 water on standing forms HCl and HOCl

Sol.:

- Due to the conversion of Cl_2 into HCl and HOCl , it loses its yellow colour.
- HOCl so formed gives nascent oxygen which is responsible for bleaching properties of Cl_2 .

65. Answer (4)

Hint: Numbering should be done in such a way that double bond and substituents get lowest number possible



2,6- Dimethyldec -3-ene

66. Answer (3)

Hint: Halogenation of alkanes is supposed to proceed via free radical mechanism.

Sol.: Rate of reaction of alkanes with halogens is $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$

67. Answer (1)

Hint: $v = \frac{c}{\lambda}$

Sol.: $\lambda = 5800 \text{ \AA} = 5800 \times 10^{-10} \text{ m}$

$$v = \frac{c}{\lambda} = \frac{3 \times 10^8 \text{ ms}^{-1}}{5800 \times 10^{-10} \text{ m}} = 5.172 \times 10^{14} \text{ s}^{-1} = 5.172 \times 10^{14} \text{ Hz}$$

68. Answer (2)

Hint: For the salt of weak acid and weak base

$$\text{pH} = 7 - \frac{\text{pK}_b}{2} + \frac{\text{pK}_a}{2}$$

Sol.: $\text{pK}_a = 3.74$, $\text{pK}_b = 4.75$

$$\text{pH} = 7 - \frac{4.75}{2} + \frac{3.74}{2} = 6.49$$

69. Answer (1)

Hint: $\Delta T_f = iK_f.m$

Sol.: $(\Delta T_f)_{\text{NaCl}} = 2 \times K_f \times \frac{0.1}{250} \times 1000 = \frac{4}{5} K_f$

$$(\Delta T_f)_{\text{CaCl}_2} = 3 \times K_f \times \frac{0.1}{250} \times 1000 = \frac{6}{5} K_f$$

$$(\Delta T_f)_{\text{Ca}_3(\text{PO}_4)_2} = 5 \times K_f \times \frac{0.1}{250} \times 1000 = 2K_f$$

$$T_1 = (T_f)_{\text{NaCl}} = -(\Delta T_f)_{\text{NaCl}} = -\frac{4}{5} K_f$$

$$T_2 = (T_f)_{\text{CaCl}_2} = -(\Delta T_f)_{\text{CaCl}_2} = -\frac{6}{5} K_f$$

$$T_3 = (T_f)_{\text{Ca}_3(\text{PO}_4)_2} = -(\Delta T_f)_{\text{Ca}_3(\text{PO}_4)_2} = -2K_f$$

Freezing point order : $T_1 > T_2 > T_3$

70. Answer (2)

Hint: In end-centred unit cell, constituent particles are present at all the corners and at centre of any two opposite faces.

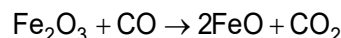
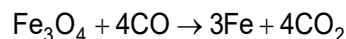
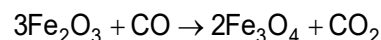
Sol.: Orthorhombic and monoclinic systems contain end-centred unit cells.

71. Answer (4)

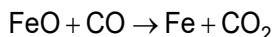
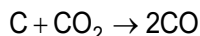
Hint: Higher temperature range of blast furnace is 900-1500 K.

Sol.:

- At 500-800 K (lower temperature range in the blast furnace)



- At 900-1500 K (Higher temperature range in the blast furnace)



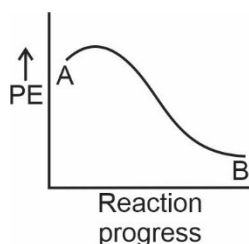
72. Answer (2)

Hint: $\Delta H = (E_a)_f - (E_a)_b$

Sol.: $\Delta H = 4.2 - 6.4 = -2.2 \text{ kJ mol}^{-1}$

$\therefore$ Given reaction is exothermic

$\therefore$ In exothermic reaction, potential energy of reactant will be greater than potential energy of product



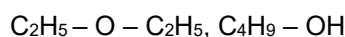
73. Answer (4)

Hint: Alcohols and ethers are functional group isomers of each other

Sol.:

- CH_4O : CH_3OH only

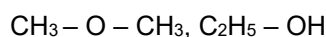
- $\text{C}_4\text{H}_{10}\text{O}$:



- $\text{C}_3\text{H}_8\text{O}$:



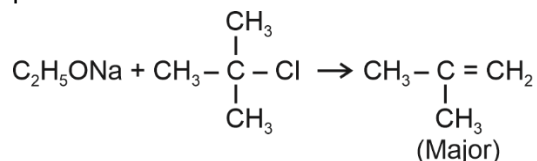
- $\text{C}_2\text{H}_6\text{O}$:



74. Answer (3)

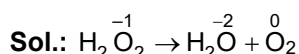
Hint: Alkoxides are strong bases along with strong nucleophile.

Sol.: The major product of the given reaction is 2-Methylprop-1-ene. Elimination reaction predominates over substitution



75. Answer (3)

Hint: Reduction and oxidation of same species simultaneously is known as disproportionation reaction.



76. Answer (2)

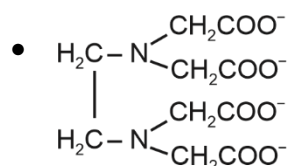
Hint & Sol: Chloramphenicol is a broad spectrum antibiotic while penicillin G has a narrow spectrum.

77. Answer (1)

Hint: SCN^- ion is an ambidentate ligand

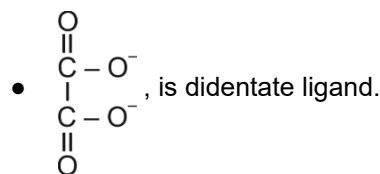
Sol.:

- SCN^- ion can coordinate through either sulphur or nitrogen atom.



(EDTA) ion is hexadentate ligand

- NO_2^- can coordinate through nitrogen or oxygen atom



78. Answer (4)

Hint: Zone refining is used when impurities are more soluble in the melt than in the solid state of the metal.

Sol.:

- Distillation is useful for low boiling metals like Zn and Hg
- Zone refining is useful for producing high purity semiconductor and other metals
- Chromatography is very useful for purification of the elements which are available in minute quantities
- Electrolytic refining can be used for Zn and Cu

79. Answer (3)

Hint: Only primary amines give carbylamine reaction

Sol.: Carbylamine reaction:

- $\text{R} - \text{NH}_2 + \text{CHCl}_3 + 3\text{KOH} \rightarrow \text{R} - \text{NC} + 3\text{KCl} + 3\text{H}_2\text{O}$
- $(\text{CH}_3)_3\text{C} - \text{NH}_2 + \text{CHCl}_3 + 3\text{KOH} \rightarrow (\text{CH}_3)_3\text{C} - \text{NC} + 3\text{KCl} + 3\text{H}_2\text{O}$

80. Answer (3)

Hint: All alkaline earth metal nitrate decompose on heating to give NO_2 gas

Sol.:

- $4\text{LiNO}_3 \xrightarrow{\Delta} 2\text{Li}_2\text{O} + 4\text{NO}_2 + \text{O}_2$
- $2\text{Ca}(\text{NO}_3)_2 \xrightarrow{\Delta} 2\text{CaO} + 4\text{NO}_2 + \text{O}_2$
- $2\text{Ba}(\text{NO}_3)_2 \xrightarrow{\Delta} 2\text{BaO} + 4\text{NO}_2 + \text{O}_2$
- $\text{KNO}_3 \xrightarrow{\Delta} \text{KNO}_2 + \frac{1}{2}\text{O}_2$

81. Answer (3)

Hint: Angle of rotation about C – C bond is known as torsional or dihedral angle.

Sol.: Of all conformations of ethane, the staggered form has maximum torsional angle and eclipsed form has least torsional angle.

Order of torsional angle : Staggered > Skew > Eclipsed

82. Answer (4)

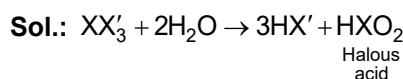
Hint: $\frac{n_C}{n_H} = \frac{1}{2} \times \frac{(V_{\text{CO}_2})_{\text{STP}}}{(V_{\text{H}_2\text{O}})_{\text{STP}}}$

Sol.: $\therefore \frac{n_C}{n_H} = \frac{11.2}{16.8} \times \frac{1}{2}$
 $= \frac{1}{3}$

$\therefore$ Empirical formula of hydrocarbon is C_1H_3 or CH_3

83. Answer (4)

Hint: In XX'_3 , the oxidation state of larger halogen (i.e. X) is +3 while of smaller halogen (i.e. X') is –1



84. Answer (3)

Hint: Tensile strength of fibres are very high

Sol.:

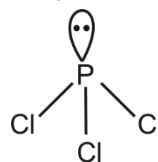
- Polythene, polystyrene and polyvinyl chloride are thermoplastic polymer
- Polyester (terylene) is a fibre

85. Answer (1)

Hint: Shape of molecule depends on total number of electron pairs present on central atom.

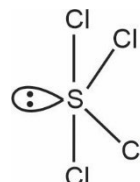
Sol.:

- PCl_3



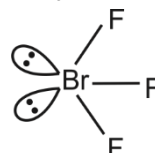
P has 3 bond pairs and 1 lone pair in PCl_3 therefore, shape of PCl_3 molecule is pyramidal

- SCl_4



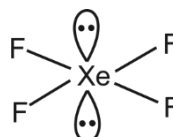
S has 4 bond pairs and 1 lone pair in SCl_4 therefore, shape of SCl_4 molecule is see-saw

- BrF_3



Br has 3 bond pairs and 2 lone pairs in BrF_3 therefore, shape of BrF_3 molecule is bent T-shaped

- XeF_4



Xe has 4 bond pairs and 2 lone pairs in XeF_4 therefore, shape of XeF_4 molecule is square planar

SECTION-B

86. Answer (4)

Hint: For spontaneous process ΔG should be negative.

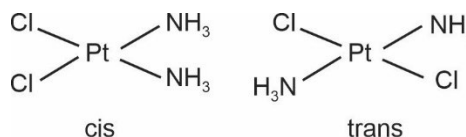
Sol.: If $\Delta H < 0$ and $\Delta S < 0$ then ΔG will be negative only at low temperature.

87. Answer (3)

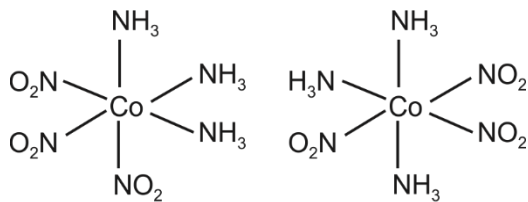
Hint: $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$ is a hydrate isomer of $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$

Sol.:

- Square planar complexes of formula MX_2L_2 (X and L are unidentate ligands) show cis and trans isomerism



- Octahedral complexes of formula $[Ma_3b_3]$ can show facial and meridional isomerism



fac-isomers

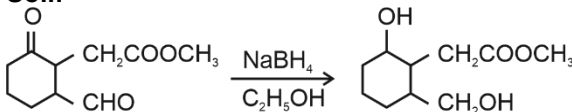
mer-isomers

- $[Co(NH_3)_5Br]SO_4$ and $[Co(NH_3)_5SO_4]Br$ are ionisation isomers of each other

88. Answer (3)

Hint: $NaBH_4$ cannot reduce ester group to alcohol

Sol.:



89. Answer (4)

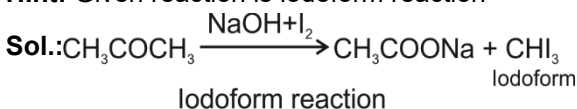
Hint: H-X bond dissociation enthalpy decreases down the group in hydrogen halide.

Sol.: Weaker is the H-X bond, stronger is the acidic character.

Acidic character order : $HF < HCl < HBr < HI$

90. Answer (2)

Hint: Given reaction is iodoform reaction



91. Answer (2)

Hint: Bond order = $\frac{1}{2}(n_{BMO} - n_{ABMO})$

Sol.:

- $O_2^+(15e^-) = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2 \left\{ \begin{matrix} \pi 2p_x^2 \\ \pi 2p_y^2 \end{matrix} \right\} \left\{ \begin{matrix} \pi^* 2p_x^1 \\ \pi^* 2p_y^0 \end{matrix} \right\}$

$$BO = \frac{1}{2}(10 - 5) = 2.5$$

- $O_2(16e^-) = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2 \left\{ \begin{matrix} \pi 2p_x^2 \\ \pi 2p_y^2 \end{matrix} \right\} \left\{ \begin{matrix} \pi^* 2p_x^1 \\ \pi^* 2p_y^1 \end{matrix} \right\}$

$$BO = \frac{1}{2}(10 - 6) = 2$$

- $O_2^-(17e^-) = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2 \left\{ \begin{matrix} \pi 2p_x^2 \\ \pi 2p_y^2 \end{matrix} \right\} \left\{ \begin{matrix} \pi^* 2p_x^2 \\ \pi^* 2p_y^1 \end{matrix} \right\}$

$$BO = \frac{1}{2}(10 - 7) = 1.5$$

- $O_2^{2-}(18e^-) = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2 \left\{ \begin{matrix} \pi 2p_x^2 \\ \pi 2p_y^2 \end{matrix} \right\} \left\{ \begin{matrix} \pi^* 2p_x^2 \\ \pi^* 2p_y^2 \end{matrix} \right\}$

$$BO = \frac{1}{2}(10 - 8) = 1$$

92. Answer (3)

Hint: Methemoglobinemia is blue baby syndrome

Sol.:

- Excessive sulphate (> 500 ppm) in drinking water causes laxative effect
- Excess lead (> 500 ppb) in drinking water can cause kidney damage
- Excess nitrate (> 50 ppm) in drinking water can cause methemoglobinemia
- Excess fluorides (> 2 ppm) in drinking water can cause brown mottling of teeth

93. Answer (2)

Hint: $E_{cell} = E_{cell}^{\circ} - \frac{0.0591}{2} \log \frac{[Mg^{2+}]}{[Cu^{2+}]}$

Sol.: $E_{cell} = 2.7 - \frac{0.0591}{2} \log \frac{(0.001)}{(0.01)} = 2.729$

$$J \ 2.73 \ V$$

94. Answer (3)

Hint: $p_{SO_2} = P_{Total} \cdot x_{SO_2}$

Sol.: $n_{SO_2} = \frac{w}{64}, n_{CH_4} = \frac{w}{16}$

$$x_{SO_2} = \frac{n_{SO_2}}{n_{SO_2} + n_{CH_4}} = \frac{\frac{w}{64}}{\frac{w}{64} + \frac{w}{16}} = \frac{1}{5}$$

$$p_{SO_2} = P_{Total} \cdot x_{SO_2} = 10 \times \frac{1}{5} = 2 \text{ atm}$$

95. Answer (2)

Hint: $P_A = P_A^{\circ} x_A$

$$P_B = P_B^{\circ} x_B$$

Sol.: $P_A = P_A^{\circ} x_A = P_B = P_B^{\circ} x_B$

$$\frac{x_A}{x_B} = \frac{P_B^{\circ}}{P_A^{\circ}}$$

$$\frac{n_A}{n_B} = \frac{P_B^{\circ}}{P_A^{\circ}} = \frac{100}{250} = \frac{2}{5}$$

96. Answer (2)

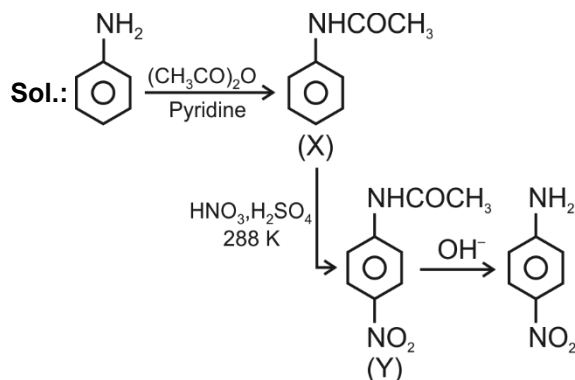
Hint: For isoelectronic ionic species, ionic radii decreases with increase in atomic numbers.

Sol.: Ionic radii : $O^{2-} > F^- > Mg^{2+} > Al^{3+}$

Greater the negative charge, larger is the ionic radii.

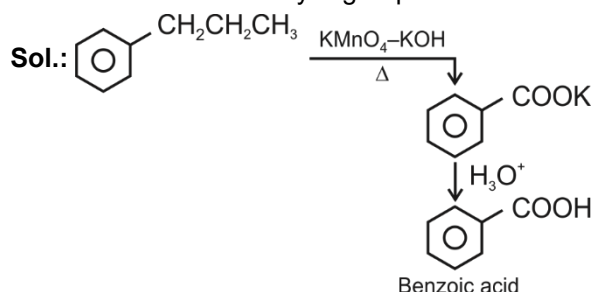
97. Answer (2)

Hint: Protection of $-NH_2$ group is done by acetylation reaction



98. Answer (3)

Hint: In vigorous oxidation of alkyl benzene with alkaline potassium permanganate, the entire chain is oxidised to the carboxylic group



99. Answer (1)

Hint: $k = Ae^{-E_a/RT}$

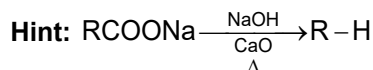
Sol.: $k = Ae^{-E_a/RT}$

$$\ln k = \ln A - \frac{E_a}{R} \left(\frac{1}{T} \right)$$

$$Y = C + mX$$

$$\therefore \text{slope } m = \frac{-E_a}{R} \text{ for plot of } \ln k \text{ vs } \frac{1}{T}$$

100. Answer (1)

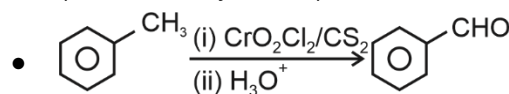


Decarboxylation reaction

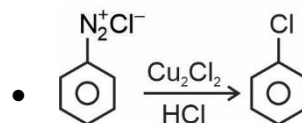
Sol.:



(Williamson synthesis)



(Etard reaction)



(Sandmeyer reaction)

[BOTANY]

SECTION-A

101. Answer (3)

Hint: Constricted pod is a recessive trait.

Sol.: Recessive traits express only in homozygous condition.

102. Answer (1)

Hint: Topoisomerase is an enzyme, employed during replication of DNA.

Sol.: 5SrRNA is synthesised by RNA polymerase III.

103. Answer (3)

Hint: Mitochondria is double membrane bound organelle and it is found in eukaryotes.

Sol.: Oxysomes are present in mitochondria. SER helps in steroid synthesis.

104. Answer (3)

Hint: During Metaphase-I, bivalent chromosomes align themselves on the equatorial plates.

Sol.: Double metaphasic plates are formed in Metaphase-I.

105. Answer (4)

Hint: Bt toxin is produced by *Bacillus thuringiensis*.

Sol.: *B. thuringiensis* is of insecticidal application.

106. Answer (2)

Hint: Habitat loss and fragmentation is the major cause driving animals and plants to extinction.

Sol.: Passenger pigeon became extinct mainly due to overexploitation.

107. Answer (2)

Hint: Diatoms are chief producers of oceans.

Sol.: Cell wall of diatoms is impregnated with silica.

108. Answer (3)

Hint: Deuteromycetes are known as 'imperfect fungi'.

Sol.: *Puccinia* and *Ustilago* belong to basidiomycetes. *Claviceps* is a sac-fungi. *Trichoderma* is a member of deuteromycetes.

109. Answer (4)

Hint: Sapindales is the order which includes the members of family Anacardiaceae.

Sol.: Convolvulaceae and Solanaceae are included in order Polymoniales.

110. Answer (2)

Hint: Herbarium is used as quick referral system for plants.

Sol.: Museum has collection of preserved plants and animals.

111. Answer (2)

Hint: Leaves in guava arise in pair at each node and lie opposite to each other.

Sol.: China rose and mustard show alternate phyllotaxy.

112. Answer (3)

Hint: Scutellum is a large, shield shaped cotyledon.

Sol.: Scutellum is mainly seen in monocot seeds.

113. Answer (1)

Hint: Parenchyma helps in photosynthesis, storage and secretion.

Sol.: Sclerenchyma consists of long and narrow cells with thick and lignified cell wall.

114. Answer (3)

Hint: Parenchyma forms major part of various organs in plants.

Sol.: Companion cells are present in angiosperms.

115. Answer (3)

Hint: Phellogen is called cork cambium.

Sol.: Secondary cortex is also known as phelloderm.

116. Answer (3)

Hint: Chlorophyceae has chl *a* and *b* as major pigments.

Sol.: Members of Rhodophyceae store food in the form of floridean starch.

117. Answer (2)

Sol.: *Selaginella* is a member of class Lycopsidea.

118. Answer (1)

Hint: Pollen grains are found in gymnosperms and angiosperms.

Sol.: All gymnosperms and angiosperms have diplontic life cycle. *Fucus* also shows diplontic life cycle.

119. Answer (3)

Hint: Facilitated diffusion requires special membrane proteins.

Sol.: Facilitated diffusion occurs along the concentration gradient.

120. Answer (2)

Hint: Breakdown of detritus by detritivores into smaller particles is known as fragmentation.

Sol.: Catabolism is carried out by saprotrophic bacteria and fungi. They secrete digestive enzymes over the fragmented detritus. The enzyme change complex organic compound into simple compound.

121. Answer (3)

Hint: *Rhizobium* is a nitrogen fixing bacteria.

Sol.:

- *Nitrosomonas/Nitrococcus* oxidises ammonia to nitrites.
- *Nitrobacter/Nitrocystis* oxidises nitrite to nitrate.

122. Answer (4)

Hint: *Rhizobium* forms nodules in leguminous plants

Sol.: *Frankia* forms nodules in non-legume plants such as *Alnus*.

123. Answer (3)

Hint: $\text{NADH} + \text{H}^+$ is formed during conversion of isocitrate to oxalosuccinic acid.

Sol.: GTP is formed during the conversion of Succinyl-CoA to succinic acid.

124. Answer (2)

Hint: Cytochrome 'C' oxidase is complex IV.

Sol.: Oxygen receives the electrons from complex IV and combines with protons to form metabolic water.

125. Answer (2)

Hint.: Jan Ingenhousz showed that in presence of sunlight it is only the green parts of the plants that could release oxygen.

Sol.: Jan Ingenhousz through his experiments showed that sunlight is essential for the plant process that helps to somehow purify the air fouled by breathing mouse and burning candle.

126. Answer (3)

Hint: 6 CO_2 molecules are required to form 1 glucose molecule.

Sol.: For production of 6 glucose molecules, 36 CO_2 molecules will be used.

127. Answer (3)

Hint: OAA is a 4 carbon compound.

Sol.: First stable product in C_4 plant i.e., sugarcane is OAA.

128. Answer (4)

Hint: ABA is also known as stress hormone.

Sol.: ABA induces the closure of stomata.

129. Answer (2)

Hint: In monoecious plants, ethylene may change the sex of developing flowers.

Sol.: Ethylene promotes the female flowers in cucumber.

130. Answer (2)

Hint: Life span of crocodile is 60 years.

Sol.: Life span of crow is 15 years.

131. Answer (3)

Hint: *Hydra* reproduces mainly through budding.

Sol.: *Penicillium* reproduces through conidia formation.

132. Answer (4)

Hint: Middle layers are ephemeral.

Sol.: Tapetum provides nourishment to developing pollen grains.

133. Answer (3)

Hint: During eutrophication, DO level of water decreases.

Sol.: Due to nutrient enrichment in water, ageing of lake occurs and process is called eutrophication.

134. Answer (2)

Hint: Phenylketonuria, colourblindness and thalassemia are Mendelian disorders.

Sol.: ABO blood group exemplifies multiple allelism and co-dominance both.

135. Answer (4)

Hint: Colourblindness and haemophilia are X-linked recessive disorders.

Sol.:

	X^h	X
X^c	X^cX^h	X^cX
Y	X^hY	XY

X^cX^h – Carrier female for both disorders

X^cX – Carrier female for colourblindness

X^hY – Haemophilic son

XY – Normal son

SECTION - B

136. Answer (3)

Hint: SER synthesizes the lipids and steroidal hormones.

Sol.: Nucleolus is the main site for rRNA synthesis.

137. Answer (2)

Hint: Two bases are held together by hydrogen bonds.

Sol.: N-Glycosidic bond holds together the pentose sugar and nitrogenous base.

138. Answer (4)

Hint: Endomembrane system includes ER, Golgi bodies, lysosomes and vacuole.

Sol.: Mitochondria is involved in ETS.

139. Answer (1)

Hint: Poly A tail is added to 3' end of hnRNA during post-transcriptional modification in eukaryotes.

Sol.: Poly A polymerase catalyses addition of adenylate residues at 3' end of hnRNA.

140. Answer (2)

Hint: Streptokinase has been commercialised as 'clot buster'

Sol.: Streptokinase is produced by *Streptococcus*.

141. Answer (2)

Hint: Karan Rai is a disease resistant variety of *Brassica*.

Sol.: Himgiri is a disease resistant variety of wheat.

Pusa Shubhra and Pusa Komal are disease resistant varieties of cauliflower and cowpea, respectively.

142. Answer (4)

Hint: Reproduction is not seen in sterile organisms like worker bees.

Sol.: Reproduction is a characteristic feature of living organism.

143. Answer (3)

Sol.: Thylacine got recently extinct from Australia.

144. Answer (1)

Sol.: *Mycoplasma* is the smallest living cell known and they lack cell wall.

145. Answer (2)

Hint: Regulators are able to maintain homeostasis by physiological as well as behavioural means which ensure constant body temperature and constant osmotic concentration

Sol.: Rat, human and crow are regulators.

146. Answer (2)

Hint: In competition, both of the interacting species are affected.

Sol.: In parasitism and Predation, one of the interacting species is benefitted.

147. Answer (2)

Sol.: Iron activates the catalase enzyme.

148. Answer (1)

Hint: Plants trap 1% of solar energy and only 10% of the energy of previous trophic level is received by next trophic level.

Sol.:

Producers → Grasshopper → Birds → man
1000 kcal → 100 kcal → 10 kcal → 1 kcal

149. Answer (2)

Hint: Embryo sac of angiosperms is 7 celled and 8 nucleate structure.

Sol.: Female gametophyte contains three celled egg apparatus and three antipodals at chalazal end.

150. Answer (4)

Hint: Good ozone is present in upper part of the atmosphere.

Sol.: Ozone present in troposphere is a secondary pollutant and is called bad ozone.

Snow-blindness is inflammation of cornea and it is caused due to high dose of UV-B.

[ZOOLOGY]

SECTION-A

151. Answer (2)

Hint: It is a multi-layered structure.

Sol.: During unfavorable conditions, *Amoeba*, withdraws its pseudopodia and secretes a three-layered hard covering or cyst around itself. This phenomenon is termed as encystation.

152. Answer (3)

Hint: In reptiles, males have copulatory organ.

Sol.: Organisms that exhibit external fertilisation, show great synchrony between the sexes and release a large number of gametes into the surrounding medium (water) in order to enhance the chances of syngamy, e.g., bony fishes and frogs. Reptiles, birds and mammals exhibit internal fertilisation.

153. Answer (4)

Hint: It is a non-reducing disaccharide.

Sol.: The male sex accessory glands are: seminal vesicles, prostate gland and bulbourethral glands. Secretions of these glands constitute seminal plasma, which is rich in fructose, calcium and certain enzymes.

154. Answer (1)

Hint: The first sign of a growing foetus may be noticed by listening to the heart sound.

Sol.:

First movements of foetus	–	Fifth month
Foetus develops limbs and digits	–	End of second month
Embryo's heart is formed	–	End of first month
Foetus' body is covered with fine hair, eye-lids separate and eyelashes are formed	–	End of second trimester

155. Answer (2)

Hint: Hormone synthesized from both ovary and placenta.

Sol.: The maximum concentration of estrogen is during puberty. Later on it shows a cyclic rise and fall during menstrual cycle. It helps in growth and maturation of follicles. Prolactin is also released in non-pregnant females. hCG, hPL and relaxin are released only during pregnancy.

156. Answer (3)

Hint: LH and FSH are secreted by pituitary gland.

Sol.: During follicular phase of menstrual cycle, the granulosa cells of growing follicles secrete estrogen. Increased secretion of oestrogen induces production of GnRH by hypothalamus. This in turn induces secretion of LH from pituitary, leading to LH surge and rupture of Graafian follicle to release secondary oocyte (ovulation).

157. Answer (3)

Hint: Mechanism of action is similar to progestasert.

Sol.: Hormone releasing IUDs like progestasert, LNG-20 increase phagocytosis of sperms, make the uterus unsuitable for implantation and the cervix hostile to the sperms.

Non-medicated IUDs like Lippes loop increase phagocytosis of sperms within the uterus.

Copper releasing IUDs like CuT, Cu7 and Multiload 375 release Cu ions which suppress sperm motility and the fertilizing capacity of sperms.

158. Answer (3)

Hint: Natural method of contraception not included

Sol.: Complications associated with STIs due to delay in treatment – infertility, ectopic pregnancy, abortion, still birth, cancer of reproductive tract.

Periodic abstinence – It is a natural method of contraception.

MTP (medical termination of pregnancy) – It is the intentional or voluntary termination of pregnancy before full term.

159. Answer (2)

Hint: In this, fertilisation occurs *in vivo*.

Sol.: ZIFT (zygote intra fallopian transfer) – In this, the zygote or early embryos (up to 8 blastomeres) are transferred into the fallopian tube.

AI (artificial insemination) – In this, semen collected from husband or a healthy donor is artificially introduced in the vagina or uterus of female and is used in infertility cases in which male has very low sperm count or inability of male partner to inseminate female.

ICSI (intra cytoplasmic sperm injection) – Sperm is directly injected into the ovum.

IUT (intra uterine transfer) – Embryos with more than 8 blastomeres are transferred into the uterus.

160. Answer (2)

Hint: Many varieties of finches were found on the same island.

Sol.: The process of evolution of different species in a given geographical area starting from a point and literally radiating to other areas of geography (habitats) is called adaptive radiation.

The founder's effect is the reduction in genetic variations that results when a small subset of a large population establishes a new colony.

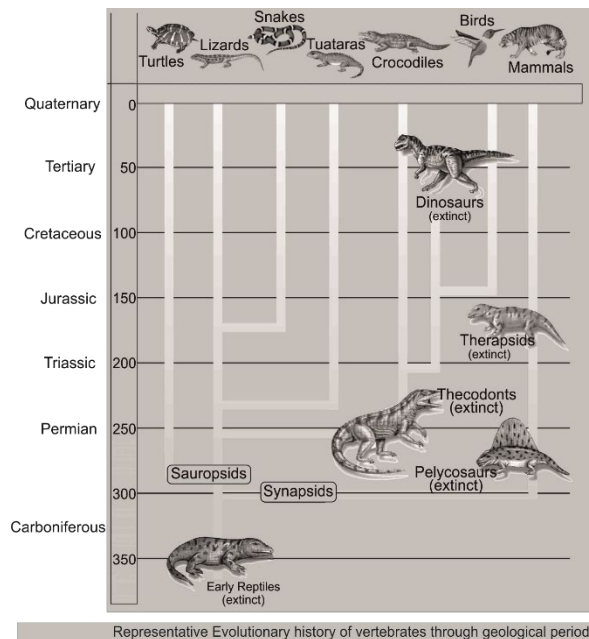
Darwin's finches are great examples of adaptive radiation and founder's effect.

The original ancestors of Darwin's finches were seed eating.

161. Answer (4)

Hint: They develop from Pelycosaur (extinct).

Sol.:



162. Answer (4)

Hint: Neanderthal man lived around 1,00,000-40,000 years ago

Sol.: Neanderthal man: Had a brain size of around 1400 cc, lived near east and central Asia between 1,00,000 – 40,000 years back.

Australopithecines: Probably lived in East African grasslands, hunted with stone weapons, but essentially ate fruits.

163. Answer (3)

Hint: Ionizing radiations have enough energy to remove an electron from an atom and can cause chemical changes in cells.

Sol.: Carcinogens are physical, chemical or biological agents that can cause transformation of normal cells into cancerous neoplastic cells.

Physical carcinogens: Ionizing radiations like X-rays and gamma rays and non-ionising radiations like UV rays.

Biological carcinogens: Several genes called oncogenes or proto-oncogenes in normal cells which, when activated under certain conditions, could lead to oncogenic transformation of cells.

164. Answer (4)

Hint: HIV is a retrovirus.

Sol.: HIV is a retrovirus, having an envelope enclosing two identical molecules of ssRNA genome and two copies of enzyme reverse transcriptase.

HIV uses CD4 receptors to gain entry into helper T-lymphocytes. HIV infection can spread from infected mother to the unborn child through placenta.

165. Answer (3)

Hint: They prefer larger, slow moving bodies of freshwater, with soft muddy bottoms.

Sol.: Freshwater fishes – *Catla*, *Rohu* and common carp.

Marine Fishes – *Hilsa*, Sardines, Mackerel and Pomfrets.

166. Answer (2)

Hint: Hormone FSH induces superovulation.

Sol.: In MOET (multiple ovulation embryo transfer technology), female animal is administered hormones with FSH-like activity which induce superovulation and produce 6-8 eggs. The fertilised eggs at 8-32 cells stages, are recovered non-surgically and transferred to surrogate mothers.

167. Answer (3)

Hint: A heterologous host is the organism in which such a gene is expressed that is not naturally present.

Sol.: Expressed sequence tags – It is an approach in which we focus on identifying all the genes that are expressed as RNA.

Gene expression – In this, a gene forms a copy of functional mRNA and then a functional protein.

168. Answer (4)

Hint: Inactivation of selectable marker

Sol.: Heat shock method – Bacterial cells are incubated with rDNA on ice, followed by placing them briefly at 42°C and then putting them back on ice. This enables bacteria to take up the rDNA.

Microinjection is mainly used in animal cells, and gene gun method is mainly used for plant cells.

Insertional inactivation is a method to differentiate recombinants from non-recombinants on the basis of their ability to produce colour in the presence of a chromogenic substrate.

169. Answer (3)

Hint: Insect gut pH is similar to human intestinal pH.

Sol.: Once the insect ingests, the plant containing inactive Bt protoxins, the alkaline pH of insect gut solubilises the crystals and convert inactive protoxin to active toxin, which binds to midgut epithelial cells. It creates pores that cause swelling and lysis and eventually causes death of the insect.

170. Answer (2)

Hint: Adenosine deaminase is crucial for the functioning of immune system.

Sol.: ADA deficiency is caused due to deletion of the gene for adenosine deaminase. In some children, ADA deficiency can be cured by bone marrow transplantation and in others it can be treated by enzyme replacement therapy.

171. Answer (3)

Hint: The enzyme convert the colourless substrate into coloured product.

Sol.: RNase – It is a type of nuclease that degrades RNA.

Ribozymes – They are nucleic acids that act as enzymes.

Alkaline phosphatase – It is an enzyme conjugated with antibody in ELISA, which yields a yellow product when p-nitrophenyl phosphate (most common substrate for ALP) is added in the microtitre plate.

172. Answer (2)

Hint: Hemichordates have proboscis gland as their excretory organ.

Sol.: *Chelone* (Turtle): Reptile; kidneys are excretory structures.

Ophiura: Echinoderm; excretory system is absent.

Dentalium: Mollusc; feather like gills perform respiratory and excretory functions.

Balanoglossus and *Saccoglossus*: Hemichordates; proboscis gland acts as an excretory organ.

173. Answer (4)

Hint: Ammocoete larva undergoes metamorphosis in freshwater.

Sol.: Cyclostomes are ectoparasites on some fishes, have an elongated body bearing 6-15 pairs of gill slits for respiration. They are marine, but migrate for spawning to freshwater and die within few days after spawning. Their larvae, after metamorphosis, return to the ocean.

174. Answer (2)

Hint: Air bladder is present in Osteichthyes.

Sol.: *Pleurobrachia* is a ctenophore which shows the property of bioluminescence (property of a living organism to emit light).

Obelia is a coelenterate which exhibits alternation of generation (metagenesis).

Pila is a mollusc, whose body is unsegmented with a distinct head, muscular foot and visceral hump.

Carcharodon (great white shark) belongs to class Chondrichthyes, having placoid scales and air bladder is absent.

175. Answer (3)

Hint: These junctions help in keeping the neighbouring cells together.

Sol.: Tight junctions – Help to stop substances from leaking across a tissue.

Adhering junctions – Perform cementing to keep neighbouring cells together.

Gap junctions – Facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells.

176. Answer (3)

Hint: Majority of neural tissue is constituted by neuroglial cells.

Sol.: Neuroglial cells constitute more than one half the volume of neural tissue in human body. Their main function is to protect and support neurons.

177. Answer (3)

Hint: Lectin

Sol.: Primary metabolites have identifiable functions and play known roles in normal physiological processes, e.g. glucose, tryptophan (amino acid), glycerol etc.

Secondary metabolites functions are not properly known. Some of them are ecologically important and some are useful to human welfare, e.g. cellulose is a polymeric substance, concanavalin A is a lectin.

178. Answer (1)

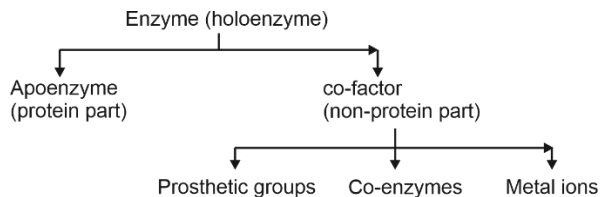
Hint: Only one type of helices are observed.

Sol.: In proteins, only right handed helices are observed. Example – keratin protein present in hair.

179. Answer (2)

Hint: Lectin

Sol.:



Metal ions act as co-factors for some enzymes. Co-factors play a crucial role in the action of enzymes, so enzymes lose catalytic activity when the co-factor is removed.

180. Answer (3)

Hint: Nucleic acids are broken down by enzymes present in the pancreatic juice.

Sol.: Nucleases are present in pancreatic juice and catalyze the following reaction:

Nucleic acid $\xrightarrow{\text{Nucleases}}$ Nucleotides $\rightarrow$ Nucleosides

Nucleotidases and nucleosidases are present in succus entericus which catalyze the following reaction:

Nucleotides $\xrightarrow{\text{Nucleotidases}}$ Nucleosides
 $\xrightarrow{\text{Nucleosidases}}$ Sugars + Bases

181. Answer (2)

Hint: They contain liver cells.

Sol.: Hepatic lobules are the structural and functional units of liver containing hepatic cells arranged in the form of cords. Glisson's capsule is a thin connective tissue sheath which covers each lobule.

Kupffer cells are phagocytic cells present in the liver.

182. Answer (2)

Hint: Succus entericus contains dipeptidases which act on the product of reaction catalysed by carboxypeptidases

Sol.: Proteins, proteoses and peptones present in chyme reaching the intestine are acted upon by the enzymes in pancreatic juice.

Proteins }
 Peptones } $\xrightarrow[\text{carboxypeptidase}]{\text{Trypsin/chymotrypsin}}$ Dipeptides
 Proteoses }

Enzymes in succus entericus digest dipeptides into amino acids.

Dipeptides $\xrightarrow{\text{Dipeptidases}}$ Amino acids

183. Answer (2)

Hint: It lifts up the ribs and sternum.

Sol.: During inspiration:

- Diaphragm contracts, leading to an increase in the volume of thoracic chamber in antero-posterior axis.
- External intercostal muscles contract, leading to an increase in the volume of thoracic chamber in dorso-ventral axis.

184. Answer (1)

Hint: Partial pressure of O₂ in alveoli is higher than that in oxygenated blood.

Sol.: pO₂ in alveoli – 104 mm Hg

pO₂ in oxygenated blood – 95 mm Hg

pCO₂ in deoxygenated blood – 45 mm Hg

pCO₂ in oxygenated blood – 40 mm Hg

185. Answer (4)

Hint: Stroke volume is the volume of blood pumped by each ventricle during a cardiac cycle.

Sol.: Stroke volume = End diastolic volume – End systolic volume

$$SV = 130 - 80 = 50 \text{ mL}$$

Cardiac output = stroke volume × heart rate
(per minute)

$$5500 = 50 \times \text{heart rate}$$

$$\text{Heart beat rate} = \frac{5500}{50} = 110 / \text{minute.}$$

SECTION-B

186. Answer (3)

Hint: Individuals with blood group B would have antigen B on his/her RBCs.

Sol.:

Individual with blood group	Can donate blood to individuals with blood group (s)
A	A, AB
B	B, AB
AB	AB
O	A, B, O and AB

187. Answer (4)

Hint: Frog is ureotelic.

Sol.: Ammonotelic: Bony fishes, aquatic amphibians and aquatic insects.

Ureotelic: Mammals, terrestrial amphibians and marine fishes.

Uricotelic: Reptiles, birds, land snails and insects.

188. Answer (2)

Hint: Aldosterone also acts on this segment for reabsorption of Na⁺ and water.

Sol.: Conditional reabsorption of Na⁺ and water takes place in DCT. In PCT, nearly all essential nutrients and 70-80 percent of electrolytes are reabsorbed. Reabsorption is minimum in the ascending limb of Henle's loop.

189. Answer (3)

Hint: Myosin filaments remain same in size.

Sol.: During muscle contraction :

- Actin and myosin filaments do not shorten.
- A band contains myosin filaments, which do not decrease in size, rather remain same.
- H zone is the central part of thick filament, not overlapped by thin filaments, also decreases in size.

190. Answer (3)

Hint: They also play a role in breathing.

Sol.: 1st-7th pairs of ribs – Vertebrosteral ribs, dorsally attached to vertebrae and ventrally attached to sternum.

8th, 9th and 10th pairs of ribs – Vertebrochondral ribs, dorsally attached to vertebrae and ventrally attached to 7th pairs of ribs.

11th and 12th pairs of ribs – Vertebral ribs or floating ribs, only dorsally attached to vertebrae.

191. Answer (1)

Hint: They do not have dendrites.

Sol.: Unipolar neurons: Cell body with one axon only and are usually found in embryonic stages.

Pseudounipolar neurons: Usually found in dorsal root ganglia of spinal nerves.

Bipolar neurons: Commonly found in the retina of eye.

Multipolar neurons: Commonly found in the cerebral cortex.

192. Answer (3)

Hint: Midbrain consists of more than 2 but less than 6 optic lobes.

Sol.: The dorsal portion of the midbrain consists of mainly four round swellings (lobes) called corpora quadrigemina.

Association areas are present in cerebral cortex and are responsible for complex functions like intersensory associations, memory and communication.

Cerebral aqueduct is a canal present in midbrain.

193. Answer (2)

Hint: During darkness, Na^+ channels in photoreceptor cells remain open.

Sol.:

During darkness ↓ Na^+ channels in photoreceptor cells remain open ↓ Continuous influx of Na^+ into photoreceptor cells ↓ Depolarisation of cells	During light stimulus ↓ Na^+ channels in photoreceptor cells close ↓ No influx of Na^+ and K^+ ions are continuously coming out via K^+ channels ↓ Leading to hyperpolarisation of cells
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194. Answer (2)

Hint: It is a fat soluble hormone.

Sol.: Steroid hormones like cortisol, testosterone, estrogen can easily pass through the cell membrane and interact with intracellular receptors.

Protein, peptide or polypeptide hormones like insulin, glucagon, pituitary hormone (ACTH) and amino acid derivatives (epinephrine) interact with membrane bound receptors.

195. Answer (3)

Hint: It is one of the functions of aldosterone.

Sol.: Mineralocorticoids regulate the balance of water & electrolytes in our body, e.g. aldosterone is the main mineralocorticoid in our body.

Cortisol produces anti-inflammatory reactions and suppresses the immune response.

196. Answer (3)

Hint: It is also present in most polysaccharides.

Sol.: H-bond is present between nitrogenous bases of two nucleotides of opposite strands. The nucleoside combines with a phosphate group by an phosphoester bond to form a nucleotide.

Phosphodiester bond connects two nucleotides on the same strand of a dsDNA molecule.

197. Answer (1)

Hint: This layer also forms crypts of Lieberkuhn.

Sol.: **Serosa-** Outermost layer, made up of thin mesothelium.

Muscularis- Formed by smooth muscles.

Sub-mucosa- Formed by loose connective tissue containing nerves, blood vessels and lymph vessels.

198. Answer (3)

Hint: It is the chief neurotransmitter of para sympathetic nervous system.

Sol.: Acetylcholine is a neurotransmitter essential for memory and learning. It is decreased in concentration in patients with Alzheimer's disease.

Parkinson's disease is due to low dopamine.

Schizophrenia is due to increase in dopamine.

199. Answer (1)

Hint: Disease caused by alteration in carbohydrate metabolism.

Sol.: Addison's disease is caused by under production of hormones by adrenal cortex that alters carbohydrate metabolism causing acute weakness and fatigue.

Hypothyroidism during pregnancy causes defective development and maturation of growing baby leading to stunted growth that is, cretinism.

Hyperthyroidism causes Graves' disease.

Excessive secretion of GH in adults result in severe disfigurement called acromegaly.

200. Answer (2)

Hint: Syphilis is caused by a bacteria.

Sol.:

Syphilis	–	<i>Treponema pallidum</i>
Trichomoniasis	–	<i>Trichomonas vaginalis</i>
Chancroid	–	<i>Haemophilus ducreyi</i>
Genital warts	–	Human Papilloma virus